

Genetics Lab (section 1)

Cytogenetics : the study of the inherited diseases at chromosomal level.

Molecular genetics : the study of the inherited diseases at DNA level.

chromosome $\rightarrow$ DNA $\rightarrow$ gene (allele).

* each chromosome carry one DNA.

machines :

- incubator
- Fume hood Function: remove contamination & gases.
- Autoclave machine its temperature 121°C , pressure 100 kPa "15 min" stabilization
- Centrifuge
- Balancer
- microscopes we use the course adjustment to view the image and the fine adjustment to clear it.
- water bath

- vortex: mix the pellet and supernatant together.

molecular Genetic Lab (B344) * Miss Isbert

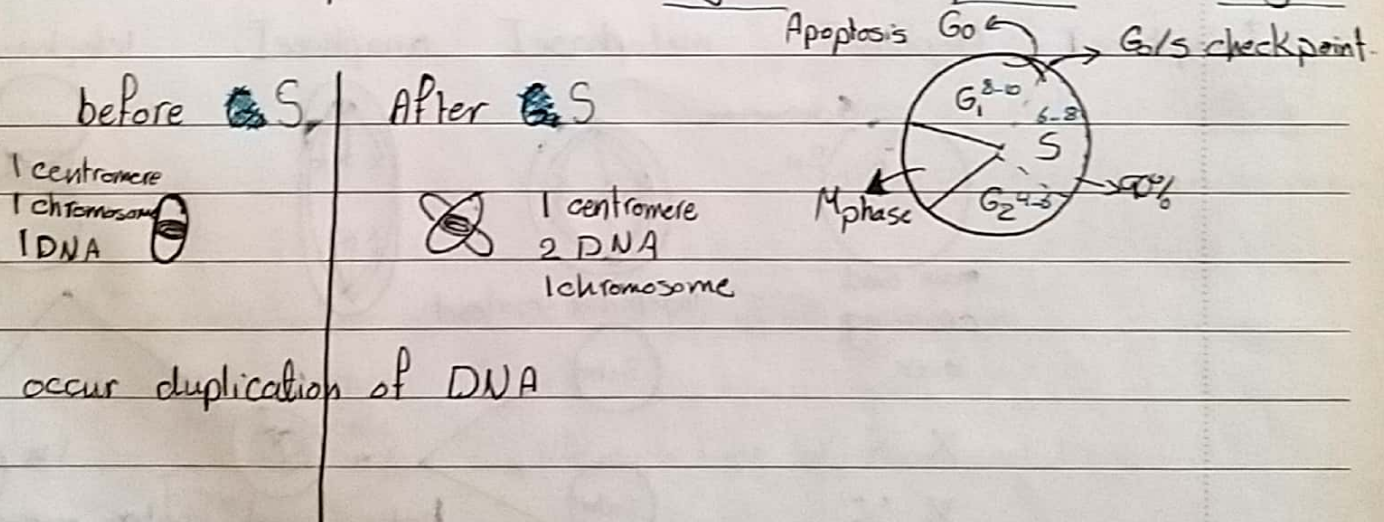
cell cycle in onion root tips

robert hook - discover the cell (in 1665)

cell theory:

- all the cell come from preexisting cells by division.
- the cell is the basic structural functional unit of all organism body
- all organism consist of one type of cell or multiple type unicellular

$G_1 \rightarrow$ duplication of all organs and protein and enzyme.



Mitosis $\rightarrow$ growth, repair, a sexual development.

Miosis $\rightarrow$ sexual development, gametes production, genetic variation

* Mitosis

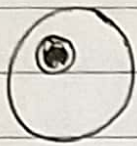
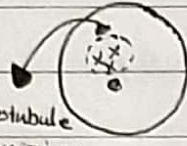
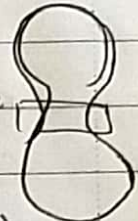
Interphase

Prophase

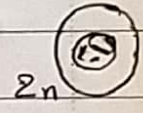
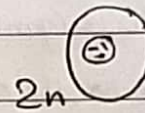
metaphase

anaphase

telophase

microtubule
organizing
centercleavage
furrow(Actin)
myosin

- 1- Kinetochore microtubule
- 2- non Kinetochore
- 3- Asters



* Meiosis (no interphase II)

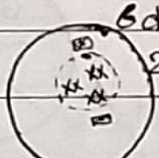
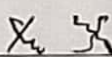
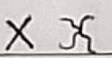
Interphase I

Prophase I

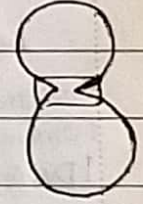
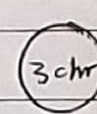
metaphase I

anaphase I

telophase I

main event
crossing
over

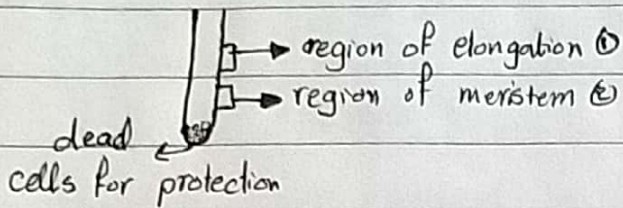
independent assortment



both must enter meiosis II



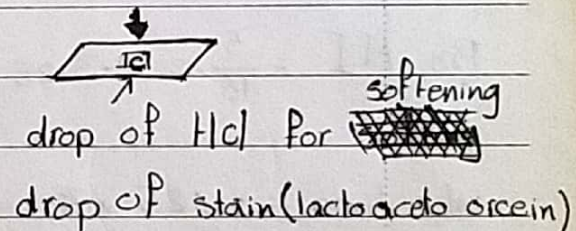
Mitosis in onion root tips.



- ① elongation without division.
- ② cell division, without elongation.

why we choose onion root tips?

- onion root tip exist in high number.
- chromosome can be easily stained and easily squashed.



MI: proliferation states in cell ~~population~~ population

$$= \frac{\text{cells in mitosis}}{\text{total \# of cells}}$$

life cycle time of onion root tips.

$$\text{Time} = \text{MI} \times 720 \text{ min}$$

$$\% \text{Time} = \frac{\text{time in each stage}}{\text{total}} \times 100\%$$

	total
Inter	10
Pro	5
meta	1
Ana	0
Telo	0
	16

$$\text{inter MI} = \frac{10}{16} = 0.625$$

$$\text{meta MI} = \frac{1}{16} = 0.0625$$

$$\text{Pro MI} = \frac{5}{16} = 0.3125$$

$$\text{Ana, telo MI} = 0$$

$$\text{inter time} = 0.625 \times 720 = 450$$

$$\text{meta time} = 0.0625 \times 720 = 45$$

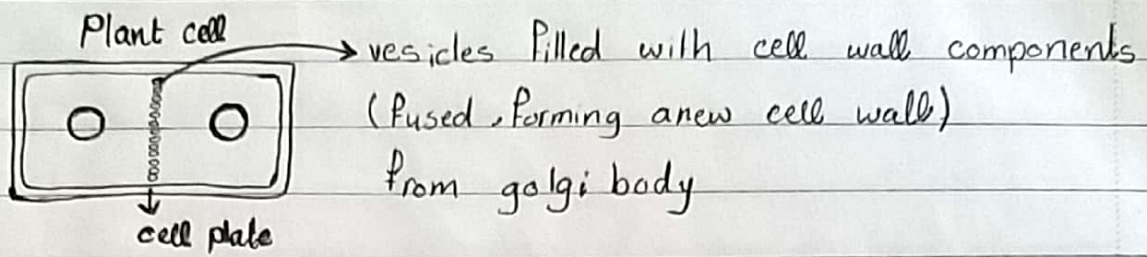
$$\text{pro time} = 0.3125 \times 720 = 225$$

$$\text{ana, telo time} = 0$$

$$\% \text{ Inter} = \frac{450}{720} \times 100 = 62\%$$

$$\% \text{ Pro} = \frac{225}{720} \times 100 = 31\%$$

$$\% \text{ meta} = \frac{45}{720} \times 100 = 6.25\%$$



there is no centriole in the plant cell, what makes the microtubules? nuclear envelope

Lab "3"

Drosophila Melanogaster

Thomas Morgan is the first scientist who discovered *Drosophila melanogaster*.

eukarya, insect, *Drosophila*, *Melanogaster*

• 3mm (size)

• short generation time (10-12) day

day 1 → female put egg 100 per day.

day 2 → larva (first instar) appear from the egg

day 3 → larva (second instar)

day 5 → larva (third instar)

day 7 → pupa

day (10-12) → flying adult (male, female)



↓ virgin (not mature)

eclosion (8 hours) (25 °C) humidity 60%

this cycle temperature dependent.

as temperature increase the cycle becomes faster.

we have a range of temperature to still *Drosophila* alive.

why Morgan choose *Drosophila*?

→ ♀ female, ♂ male

dimorphic

- male is smaller than female

♀ female

7 segment  → abdomen shape.

♂ male

 sex comb
5 segment

- small in size.
- short of cell cycle.
- easy to anesthetizing

anesthetizing methods

1) CO_2

2) ether

3) cooling

4) Flynap.

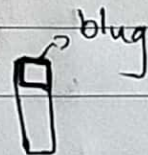
→

⊗ drosophila live in vial / bottles.

for small  population

for large Population 

morgan was use milk bottle to keep drosophila.



1g/5ml autoclaved
distilled water

9250 5051 12

N O T E B O O K

♀ ⊗ ♂

4 female : 4 male to sure that cross happened.

we put virgine female with mature male.

the female has a properites which is polyandrous.
when ♀ capable to carry more
mature than 1 egg.

females under microscope there is a black larva in the
abdomen. dark.

- we can conclude the the cross is correct
the media become less,
- or we can see a black dots which mean this is
the head.

(morgue) to Kill drosophila.

water + detergent

 then in autoclave.

drosophila have 8 chromosome

1 is sex chromosome. $X^A X^A$

2, 3, 4 autosome. AA

wt (+)

mutant

there is no relationship between (wt, mutant) and (dominant, recessive)

wild type trait:

- red eyes
- wings (long, rounded)
- body color is gray.

mutant trait:

- black body (b), II

♀ black bb

♂ wt b^+b^+, b^+b

ebony (e, II) ~~not for~~

♂ ebony ee

♀ ebony ee

♀ wt e^+e^+, e^+e

white eye (W, I)

♀ white $X^W X^W$

♀ wt $X^{W^+} X^{W^+}, X^{W^+} X^W$

♂ wt $X^{W^+} Y$

♂ white $X^W Y$

Trait	♀ female	♂ male
(st III) stubble bristles ^{شعيرات}	sbsb	$\overset{+}{s}b\overset{+}{s}b, \overset{+}{s}b\overset{+}{s}b$
(Y, I) yellow body ^{صفرة}	$X^{Y+}X^{Y+}, X^{Y+}Y$	$X^Y Y, X^{Y+} Y$
(b/I) Bar eye ^{عين مقوسة}	$X^{b+}X^{b+}, X^{b+}X^b$	$X^b Y$
(vg/II) vestigial wings ^{جناح قصير}	vgvg	$vg^{+}vg^{+}, vg^{+}vg$
(st/III) scarlet eye ^{عين حمراء}	$st^{+}st^{+}, st^{+}st$	stst
(ap/II) Apterous ^{بدون أجنحة}	apap	$ap^{+}ap^{+}, ap^{+}ap$
(Cy/II) Curly wings ^{جناح مقعرة}	$cy cy, cy^{+}cy^{+}, cy^{+}cy$	$cy^{+}cy^{+}, cy^{+}cy, cy cy$
(dp/III) dumpy wings ^{جناح مسطحة}	dpdp	$dpdp, d^{+}pdp, d^{+}pdp$
(cv/I) cross veins	$X^{cv}X^{cv}, X^{cv+}X^{cv+}, X^{cv+}X^{cv}$	$X^{cv} Y$
(ey/II) eye less	$eyey, eyey$	$eyey, ey^{+}ey^{+}, ey^{+}ey$

How eye less drosophila live?

Monohybrid, dihybrid cross.

♀ white eye (X) ♂ wt (W/I)

$X^w X^w$

(X^w)

$X^{w+} Y$

(X^{w+}) (Y)

	X^{w+}	X^{w+}	X^w
X^w	$X^{w+}X^w$	X^wX^w	
Y	$X^{w+}Y$	X^wY	

example

♂ vestigial $\otimes$ ♀ wt homo (vg/II)
 $vg\,vg$ $\times$ vg^+vg^+
 (vg) (vg⁺)

F₁ vg^+vg

F₂

	vg ⁺	vg
vg ⁺	vg ⁺ vg ⁺	vg ⁺ vg
vg	vg ⁺ vg	vgvg

3 : 1
wt mutant

♂ ebony $\otimes$ ♀ vestigial (vg/II)

$ee\,vg^+vg^+$ $e^+e^+vg\,vg$
 eug^+ e^+vg

F₁ $e^+e\,vg^+vg$ wild type.

$e^+vg^+, eug, e\,vg^+, eug$

F₂ $e^+vg^+ eug eug^+ eug$

	e^+vg^+	eug	eug^+	eug
e^+vg^+	$e^+e^+vg^+vg^+$	$e^+e\,vg^+vg$	$e\,e^+vg^+vg$	$e\,e\,vg^+vg$
eug	$e^+e\,vg^+vg$	$ee\,vg\,vg$	$ee\,vg^+vg$	$ee\,vg\,vg$
eug^+	$e\,e^+vg^+vg$	$e\,e\,vg^+vg$	$ee\,vg^+vg$	$ee\,vg\,vg$
eug	$e\,e\,vg^+vg$	$ee\,vg\,vg$	$ee\,vg^+vg$	$ee\,vg\,vg$

9 wt : 3 wt "e" : 3 wt "vg" : 1 "e" "vg"

stubble bristle $SbSb$

curly wings $CyCy$

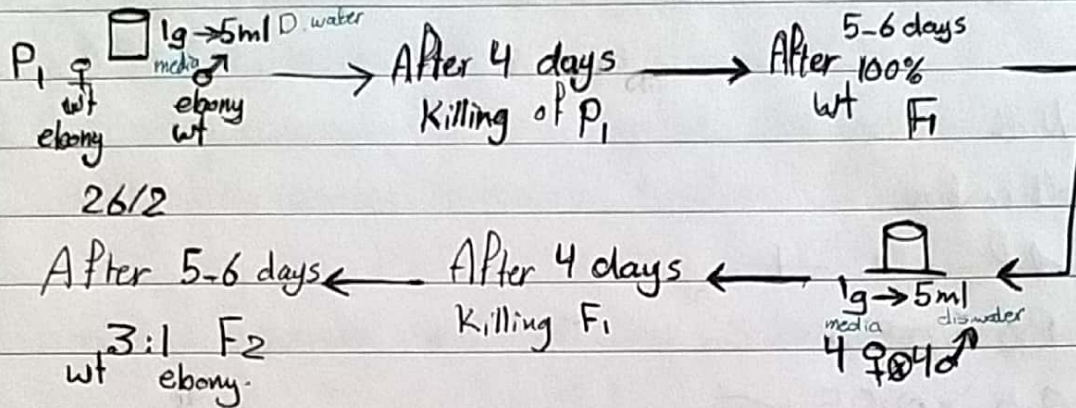
drosophila will die.

Group "2"

Lab "4"

26/2

Chi square test.



- observed value: (F_2) we see it after 5-6 days from Killing F_1 ,
- expected value: from punnet square

Null hypothesis: There is no significant diff. between E & O values, results are by chance.

Alternative hypothesis: there is a significant diff. between E & O values, results are biased

$$\chi^2_{\text{chi}} = \sum \frac{(O - E)^2}{E}$$

$$d.f. = n - 1$$

↓
degree of freedom.

$$P_{\chi^2} > P_{\text{crit}} \quad 0.05 \text{ off value}$$

$$P_{\chi^2} < 0.05$$

$$P_{\chi^2} = 0.75 \rightarrow \text{by chance accept null}$$

$$0.25 \rightarrow \text{biased reject null}$$

but not accept alternative

N O T E B O O K

	O	E	O-E	(O-E) ²	(O-E) ² /E
Head	8	10	-2	4	0.4
tail	12	10	2	4	0.4
					$\chi^2 = 0.8$

Null

alternative

$$d.f. = n - 1 = 1$$

$$P_{\chi^2} = 0.5$$

$$0.5 > 0.05 \checkmark$$

we accept the null hypothesis.

P_i: ♀ ^{homo}cy⁺cy⁺ ⊗ ♂ ^{hetero}cy⁺cy → F₁ & F₂ gen

		O	E	(O-E)	(O-E) ²	(O-E) ² /E	wt	cy	total
	♂						23	17	40
	♀						20	25	45
♂ wt		23	20	3	9	0.45			
♂ cy		17	20	-3	9	0.45			
♀ wt		20	25	-5	25	0.278			
♀ cy		25	25	0	0	0			
						$\chi^2 = 1.45$			

$$d.f. = n - 1 = 3$$

$$P_{\chi^2} = 0.75$$

$$0.75 > 0.05 \checkmark$$

accept null hypothesis

Lab "5"

cytogenetics of drosophila

(chromosomal level)

"giant chromosome"

all chromosome are giant chromosome (8 chromosome)

all chromosomes become together in the chromocenter.

* chromosomes will be in S phase with out M phase

why giant chromosome specific in larva?

because larva stage ~~that~~ need high gene expression.

giant chromosome in flying adult have a few number

A

polytene chromosome appear as a bands

dark "heterochromatin"

light "euchromatin"

puff

heterochromatin give us inactive gene expression

euchromatin give us active gene expression

puff chromatin give us hyper active gene expression

in human at different tissue or drosophila the same.

Puff is tissue specific.

we distinguished our chromosomes depends on length, band, and centromer.

the homologous chromosome in drosophila called "limb"

$\begin{bmatrix} R \\ 44 \end{bmatrix}$ Limb $\begin{cases} R \text{ right} \\ L \text{ left} \end{cases}$

name of chr: number of chromosome, Limb, Division, subdivision
(3 L 65 D)

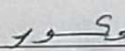
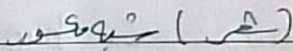
chromosome 4 very small and heterochromatin we can't see it clearly, it's position in the chromocenter.

A synapsis  branches.

construction mutation: chro. thin & elongate

sever construction: lead to breaks

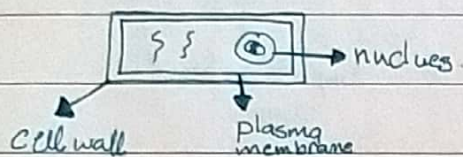
Break semibreak

polytene chromosome in larve in salivary gland.

Lab "6" DNA EXTRACTION From *D. melanogaster*

- * molecular genetics (molecular level)
- * From flying adult.
- * to extract DNA we have to degrade plasma membrane & nuclear envelope.



to extract DNA we have two methods:

• phenol-chloroform

• Salting out (Kit)

Mg^{+} → DNAase is inactive

we firstly have to: (chelating agent) ethylenediamine, tris, acetic acid.

II cell lysis

Drosophila
epidural

• we add squishing buffer (Tris, EDTA, SDS)

Denaturation for protein

maintain the PH

* cell lysis buffer

- Tris-EDTA

- SDS

- NaCl → neutralize the charge on DNA

- PK → denaturation of protein.

we add it twice to confirm the result

micropipette

blue 100 μ l - 1000 μ l عشرين 0 5 0
 yellow 10 - 100 , 20, 200 احد 0 عشرين 5
 Red 0.5 - 10 μ l

centrifuge



DNA in supernatant

denaturation of protein $\xrightarrow{\text{PC-I}}$ Antifoaming

↓
 denaturation of
 lipid & carbohydrate

[2] remove of contaminant

[3] DNA ppt

Isopropanol has more Affinity to DNA than H_2O



DNA + Isopropanol + Ethanol 70%

بطريقه كاله

24/3

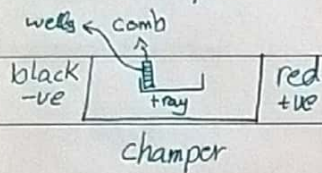
علي

26/3

نظري

Lab "7" Gel electrophoresis

After we extract DNA, to discover our DNA (Purity, size, ...) we do: gel electrophoresis

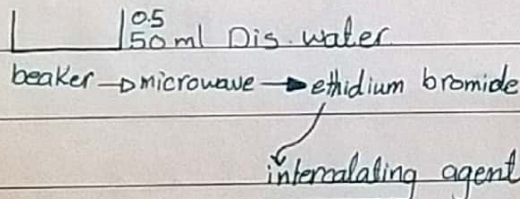


→ TBE buffer, Dis. water
Tris-borate EDTA

we choose the ~~size~~ size of chamber according to the number of sample.

the smallest chamber need 50ml agarose.

we put TBE-buffer, Dis. water, agarose in the tray.



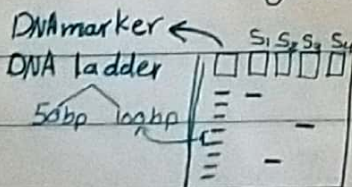
↪ between nitrogenous bases of DNA.

gel electrophoresis separate DNA according to the: size, charge, conformation



circular DNA faster than linear DNA.

Loading: put the samples in the wells



DNA (2ml) + Loading dye (2ml)

(denser material + tracking dye)

gel electrophoresis → semi-quantitation method

(if there are DNA or Not)

we can't know the concentration

nanodrop (quantitation method) ng/micro

connected to computer.

the sample size (1-2 μ L)

spectrophotometer

 diluted
Cuvette

$\lambda_{DNA} = 260$

$\lambda_{carbohydrate} = 230$

$\lambda_{protein} = 280$

① of aromatic a.a

→ His, tyr, trptophan.

conc. = $A \times 50$

$$\text{purity} = \frac{A_{260}}{A_{280}} = \square$$

(1.8 - 2)

RNA
with 

with 
Protein

$$\frac{A_{260}}{A_{230}} = \square$$

(1.8 - 2)

with carb. 

with 
protein

Lab "9" Mutation (chemical & physical mutagen)

and the frequency of mutation in population $< 1\%$

mutation: change in nucleotide sequence. single nucleotide polymorphism

what is the difference between mutation and SNP?

mutation is always correlated with diseases

SNP is a neutral change which responsible of our variation

the frequency of SNP $> 1\%$

- germline mutation: inherited

- somatic mutation: not inherited

type • Point mutation:

- deletion $ATG \rightarrow AG$

- insertion $ATG \rightarrow ATGG$

- substitution $ATG \rightarrow TTG$

silent mutation: substituted of nucleotide with another nucleotide for the same amino acid.

missense mutation: substituted of nucleotide with another nucleotide for the different amino acid.

nonsense mutation:

$ATG \rightarrow GTG$ (stop)

truncated protein (non functional)

N O T E B O O K

mutagen: physical, chemical, infectious agent, spontaneously
↳ during replication

physical: UV

X-ray, γ

non ionizing

ionizing

can't penetrate the cell.

can penetrate the cell.

λ is big, energy is low.

λ is low, energy is high.

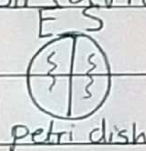
chemical: Food, cosmetics

infectious agent: integration of virus genome ^{with} human genome.

UV $\rightarrow$ mutation

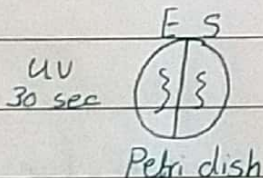
* Salmonella typhimurium (T1535) (UVr $\ominus$)

* E. coli (UVr $\oplus$)

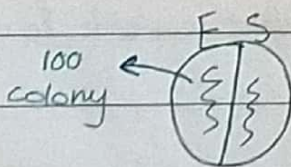


N.A
nutrient agar

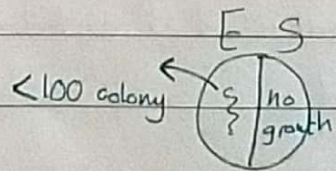
controlled.



$\Rightarrow$ incubator 37°C / 24 hr



controlled



UV

E. coli repair the mutation by its repair machine

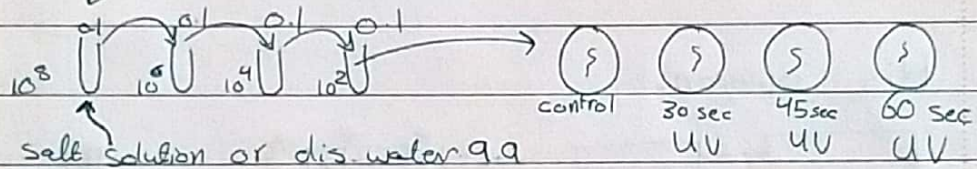
while S don't have repair machine.

N O T E B O O K

the repair machine in mutation:

- photoreactivation T=T light, photolyase
- excision (dark) T=T exonuclease, endonuclease
DNA polymerase, DNA ligase (edges) (between strands)
- Post replication recombination errors after replication.

% of UV survival E. coli



Concentration $DF = \frac{\text{Final volume}}{\text{Initial volume}}$

37 hour / 24 hour

as time with UV increase, the colony decrease

$$\% \text{ of UV survival} = \frac{\% \text{ of colony } 30 \text{ sec}}{\% \text{ of colony control}}$$